

Lecture 4

AI-CR-503

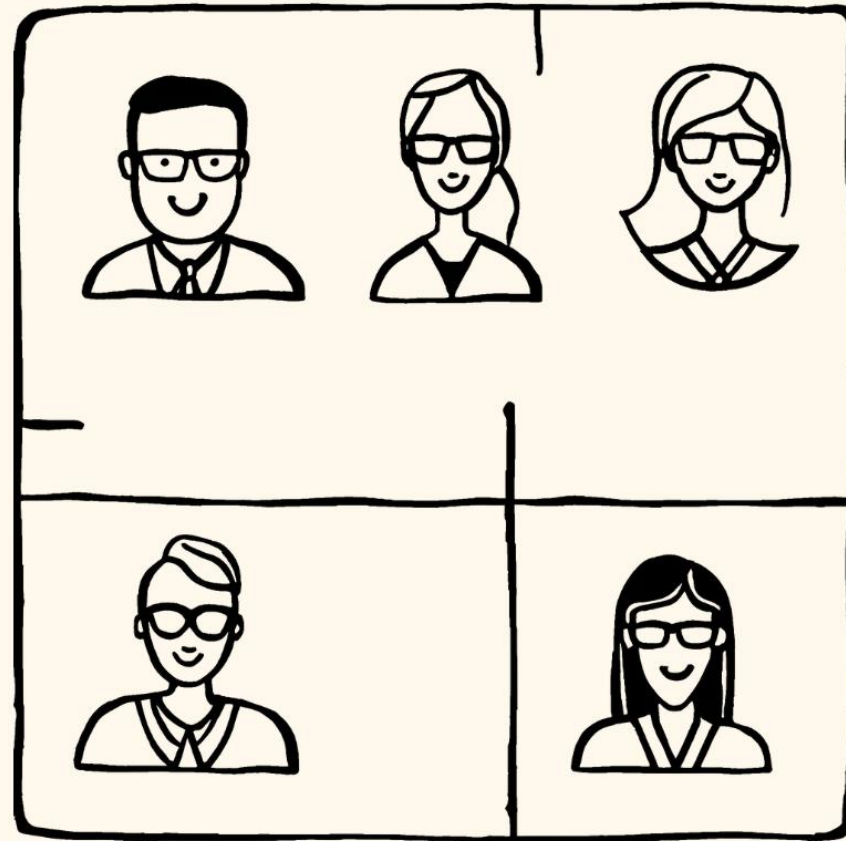
Digital Technology Program

Madan Bhandari University of Science and Technology (MBUST)

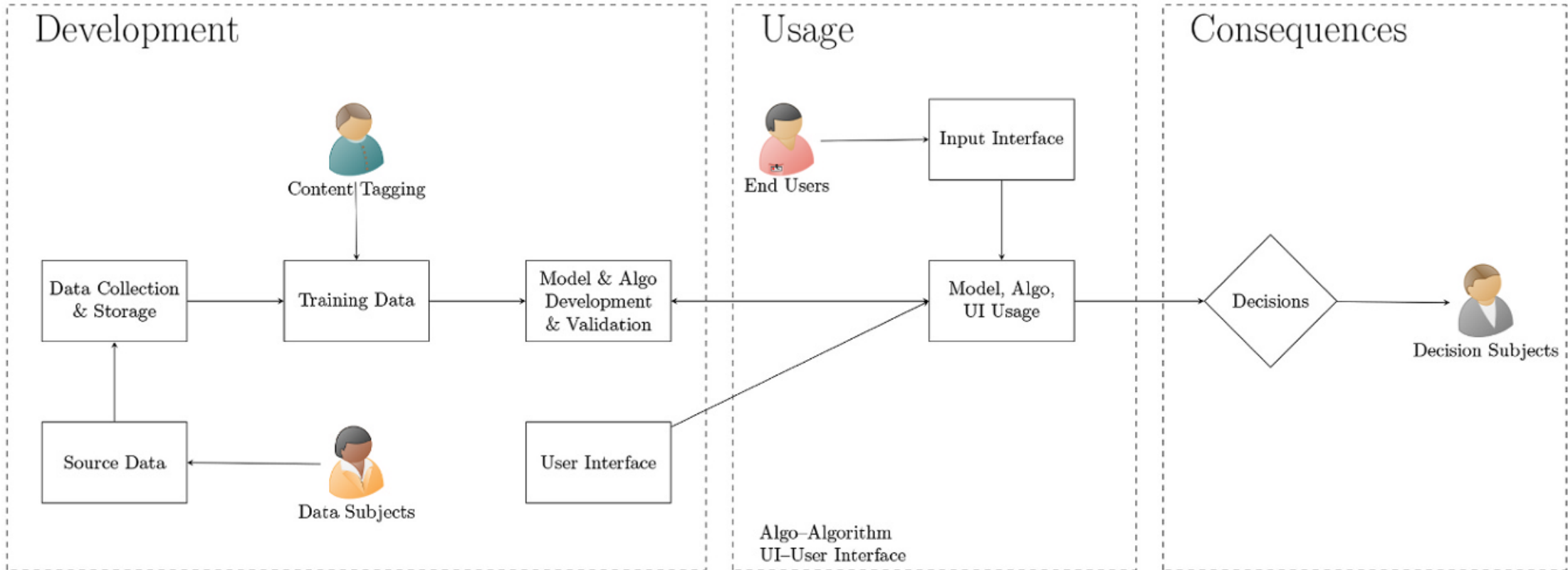
UNIT 4 Interaction with stakeholders

DESIGN-LED AI

Stakeholder Personas For AI Projects



Artificial intelligence (AI) project lifecycle



- Development: AI system development begin after planning and designing the system, data collection and processing for model training and testing.
- Usage: End user employ user interface, model, and algorithm for application output.
- Consequence: Decisions are the model or algorithm output that impacts specific people, organizations, or groups.

Who are stakeholders?

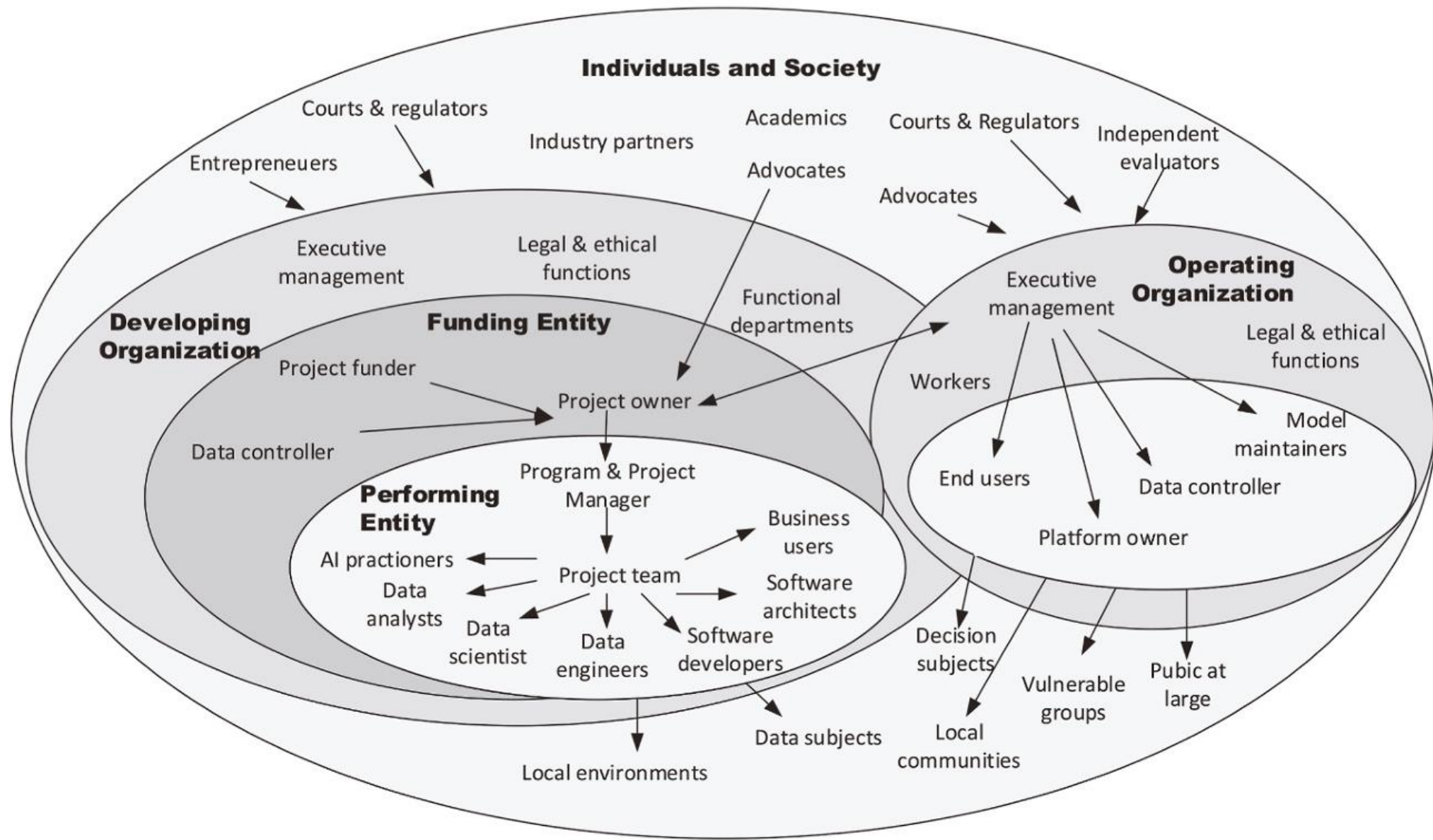
- Stakeholders are different groups from end-users to regulators that interact with AI based system for the improvement of the product as well as services.
- We use the term stakeholders to describe not only users of the AI system but also any individuals, groups, organizations, legal/regulatory institutions, and nation states which can influence or be influenced by the AI systems.

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- In this context, stakeholders interacts with AI based system that is designed using people-centred AI.
- Multiple key stakeholders can be identified by Stakeholder Mapping process.

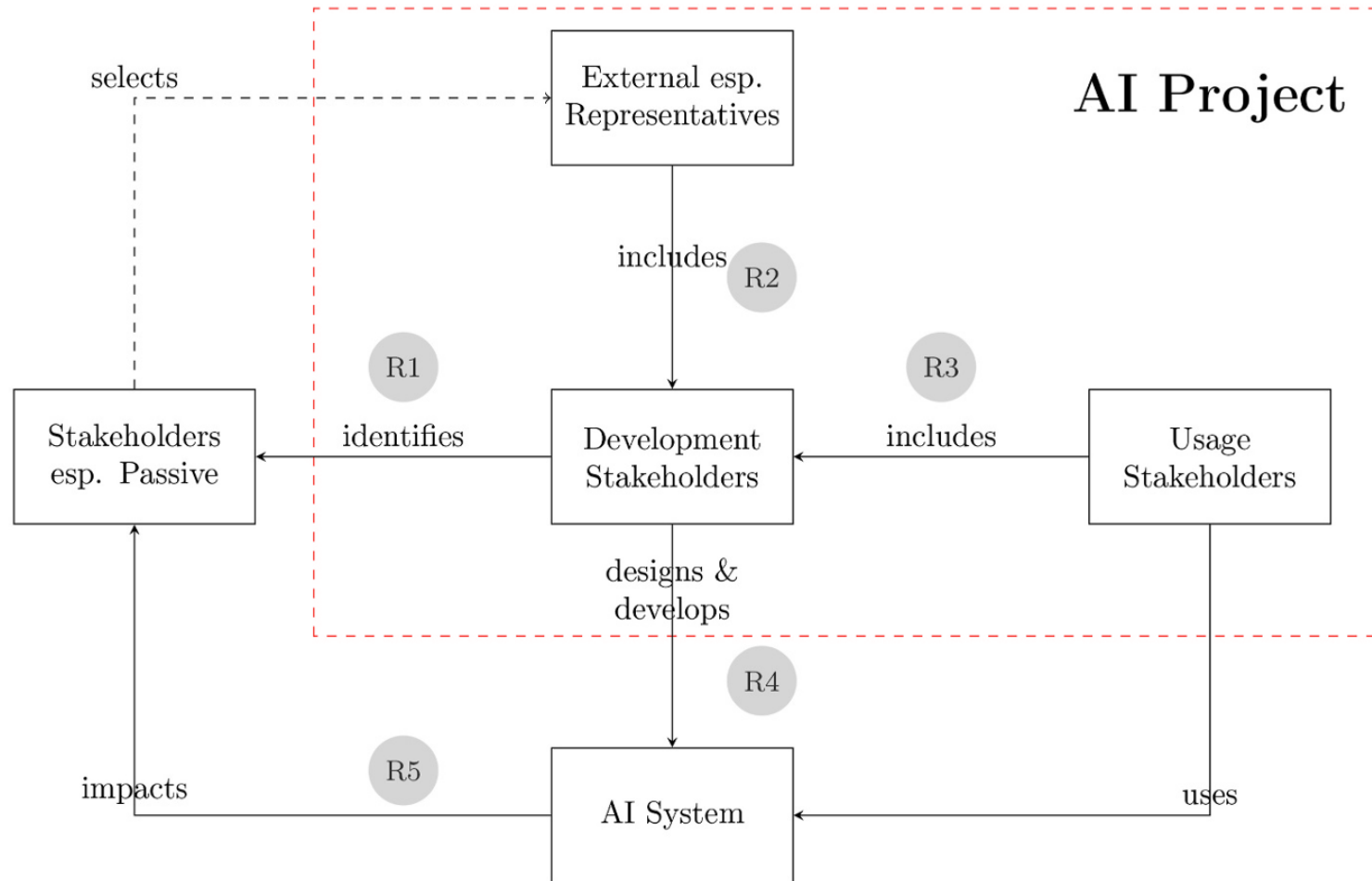
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- Three levels of stakeholders are:
 1. Individual stakeholders: users, developers, and researchers.
 2. Organizational stakeholders: NGOs, Bank, Technology company etc.
 3. National/International stakeholders: They are engaged in making laws, rules, and regulations.



Cross role relationships

AI development project stakeholders



Stakeholder participation in AI development

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- R1: AI development project stakeholders are represented.
- R2: Representatives are engaged in AI projects to balance the needs of all stakeholders.
- R3: AI projects include individuals from the operating organization in the development that can represent the interests of their stakeholders.
- R4: Combined the designers and decision-makers make important decisions on the design and development of the AI system, an artificial agent.
- R5: The operating organization is in a central role between the development stage and the consequences for users of the AI system.

Methods of interactions

- a) User-centered design: inquiring stakeholders during need finding and design evaluation.
- b) Service design: understanding and shaping value propositions and interactions among stakeholders.
- c) Participatory design (PD): mixing diverse voices and challenging power structures.
- d) Co-design: creative cooperation between stakeholders and technology designers across the whole span of a design process.
- e) Action research: improving researchers' capacities and processes.

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- f) Value-sensitive design: accounting for indirect stakeholders and moral values.
- g) Social choice theory and mechanism design: quantitative aggregation of stakeholder preferences.
- h) Participatory democracy and civic participation: involving citizens and stakeholders in a broad range of civic decision making.
- i) Deliberation theory: qualitatively weighing and discussing competing perspectives and policies.

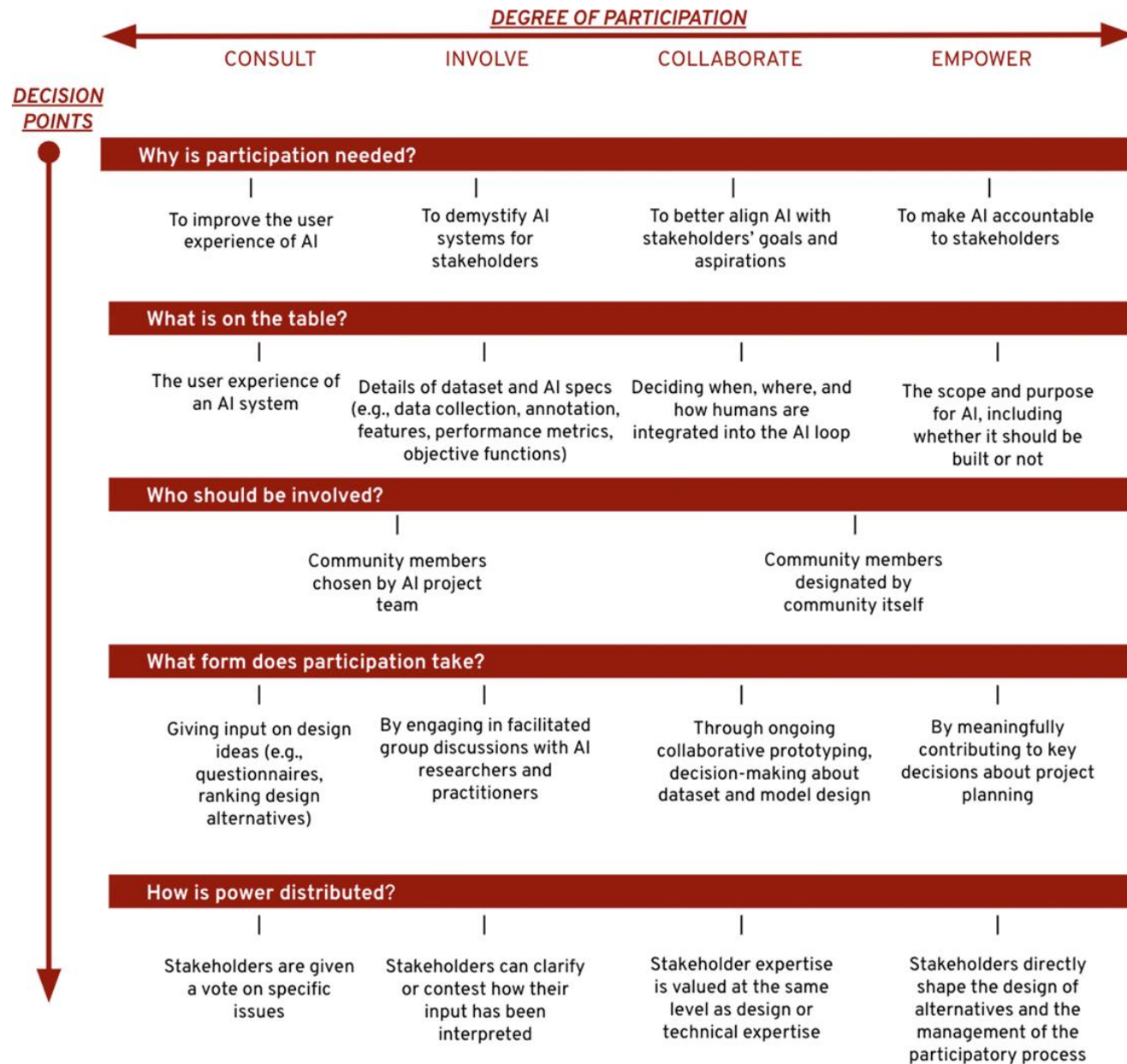
Importance of stakeholder interaction

- Power: is the extent to which the stakeholder can impose their will on the project implementation or output usage.
- Legitimacy: is evaluated when the stakeholder enjoys a socially constructed standing shared by the parties.
- Urgency: is assessed when the stakeholder claim or relationship calls for immediate attention either as a time-sensitive or critical relationship or claim.
- Harm: evaluates if the stakeholder would be harmed by the implementation or use of the system.

Increasing the stakeholder participation

- Four tactics are:

1. Consult
2. Involve
3. Collaborate
4. Empower



- Why participation is needed?
 - To improve the user experience of AI.
 - To demystify AI system for stakeholders.
 - To better align with stakeholder goals and aspirations.
 - To make AI accountable for stakeholders.

- What is on the table?

- The user experience of an AI system.
- Details of datasets and AI specs. (e.g., data collection, annotation, features, performance metrics, objective functions)
- Deciding when, where, and how humans are integrated into the AI loop.
- The scope and purpose for AI, including whether it should be built or not.

- Who should be involved?
 - Community members chosen by AI project team.
 - Community members designed by community itself.

- What form does participation take?
 - Give input on design idea (e.g., questionnaires, ranking design alternatives)
 - By engaging in facilitated group discussions with AI researchers and practitioners.
 - Through ongoing collaborative prototyping, decision making about datasets and model design.
 - By meaningfully contributing to key decisions about project planning.

- How is power distributed?
 - Stakeholders are given a vote on specific issues.
 - Stakeholders can clarify or contest how their input has been interpreted.
 - Stakeholders' expertise is valued at the same level as design or technical expertise.
 - Stakeholders directly shape the design of alternatives and the management of the participatory process.

Challenges

1. The stakeholders beyond the immediate users of the system are likely to be impacted by the AI system (either positively or negatively).
2. The AI system builders may also lack the required knowledge or the power to make decisions should the wider stakeholder ecosystem beyond the end-user of the system be impacted.

Thank you.